

Indian Statistical Institute, Bangalore

B.Math (H) Third Year

End Semester Economics I

Maximum marks: 50

Time: 3 hours

Date: November 6, 2025

Section I: 10 marks. All questions are compulsory.

Q1. If the marginal propensity to save increases, the Keynesian income/spending multiplier

a) falls b) rises c) remains the same d) cannot comment

Q2. What is Okuns law?

Q3. Consider the quantity theory of money equation. Suppose the actual money supply is M^* which is not equal to M ; according to the neutrality principle, the variable that would adjust to achieve equilibrium is:

a) V b) Y c) P d) M

Q4. State one difference between nominal and real GDP.

Q5. The depreciation of a country's real exchange rate would increase the export of goods, and make the economy more competitive. True or False?

Q6. The monetary base is:

a) The sum of currency in circulation and commercial bank reserves.

b) The foreign exchange reserve held by the central bank.

c) The official reserves held by commercial banks.

d) The sum of currency in circulation and the currency held by commercial banks.

Q7. Child care provided by a licensed day care centre is not included in GDP calculation. True or False?

Q8. Budget surplus refers to:

a) An excess of government outlays over taxes.

- b) An excess of government revenues over total outlays.
- c) The difference between fiscal deficit and interest payments.
- d) The sum of total expenditure and borrowings.

Q 9. Which of the following are not assets of a commercial bank?

- a) Government bonds
- b) Reserves
- c) Deposits with the Central Bank
- d) Savings deposits

Q10. Which one of the following is not an example of government transfer payments to individuals or firms?

- a. Social security payments
- b. Unemployment compensation
- c. Interest payments to holders of government bonds
- d. Investment in infrastructure

Section II: 30 marks. Each question carries 2 marks. Attempt any 15 questions.

Q1. What is meant by sticky wages? Give one explanation.

Q2. Distinguish between structural and cyclical unemployment.

Q3. What is the wage productivity gap?

Q4. Does rising inventories imply that an economy is slowing or growing faster than expected? Justify your answer.

Q5. Define M1 and M2. What types of monetary policy instruments can the RBI use to affect M1 and M2.

Q6. Refer to the table below.

- a) What is the revenue deficit?

b) What is the fiscal deficit?

Table: Union budget, India, (in crores of rupees), 2023-24

1	Revenue Receipts	2632281
2	Total Capital Receipts, of which	1870816
3	Recoveries of Loans and Advances	23000
4	Other Receipts (Disinvestment, etc.)	61000
5	Borrowings and Other Liabilities	1786816
6	Total Receipts	4503097
7	Total Expenditure, of which	4503097
8	Revenue Expenditure	3502136

Q7. Define the unemployment rate. What is the unemployment rate in a country with an adult population of 140 crores, labour force of 78 crores and 58 crore employed persons.

Q8. Explain the impact of the oil-price shock for an economy that is already in a recession, with a diagram.

Q9. Explain if the following would be included in the calculation of GDP or not.

- a. The rise in life expectancy over time.
- b. The semi-conductor chip that goes into the motherboard that goes into a laptop bought by a consumer.
- c. The sale of a new automobile.
- d. Salaries paid to Bangladeshi workers working in India.

Q10. What are the two main accounts in the balance-of-payment account, and how do they differ?

Q11. Of the three sectors in the economy, which has the highest share of GDP and which has the lowest in recent times.

Q12. Write two functions of money and explain each.

Q13. Write down a difference between efficiency wage and the minimum wage in an economy.

Q14. For each of the following, indicate which curve in the AS/AD model shifts (initially), and in which direction(s):

- a. An adverse supply shock
- b. An increase in government spending

Q15. Suppose the Central Bank makes an open market purchase of bonds, what would happen to the following?

- a. federal funds rate.
- b. aggregate demand in the economy.

Q16. What are the two axes of the Philips curve? Under what conditions can stagflation occur?

Q17. In the simple Keynesian model, what role does inventory investment play in achieving equilibrium, and how does it adjust when planned aggregate expenditure differs from output.

Q18. Using the Quantity Theory of Money ($MV = PY$), explain how demonetisation in India in 2016 affected the money supply (M) and, consequently, the price level (P) in the short run.

Section III: 10 marks. Each question carries 5 marks. Attempt any 2 questions.

Q1. Consider an economy with the following IS and LM curves:

$$Y = 4350 - 800r + 2G - T \text{ (IS)}$$

$$M/P = 0.5Y - 200r \text{ (LM)}$$

Suppose that $T = G = 450$ and that $M = 9000$. Find an equation for the aggregate demand curve.

Q2. The consumption, desired investment, and government spending in a closed economy are:

$$C = 360 - 200r + 0.1Y$$

$$I_d = 120 - 400r$$

$$G = 120$$

a. Find an equation for saving, S , in terms of output Y and the real interest rate r . What value of the real interest rate clears the goods market when $Y = 550$? When $Y = 600$? When $Y = 650$? Use the goods market equilibrium condition to derive the IS curve. Graph the IS curve.

In the same economy, the real money demand function is,

$$M_d / P = 100 + 0.2Y - 2000i$$

Assume that $M = 300$, $P = 2$, and $\pi_e = 0$, where, i is the nominal interest rate, and π_e is the expected inflation rate. What is the real interest rate r that clears the money market when $Y = 550$?

Or

Consider the following economy:

$$C = 200 + 0.5Y - 500r$$

$$I_d = 200 - 500r$$

$$L = 0.5Y - 250r$$

$$G = 150$$

$$M = 4900$$

$$Y = 1000$$

- What are the general equilibrium levels of the real interest rate r , the price level P , aggregate consumption C , and desired investment I_d ?
- Suppose there is a beneficial supply shock and the full-employment level of output Y increases temporarily to 1050. Show what happens to the economy in a graph. What will be the new long-run equilibrium value of r ? What is the new price level P ?

Q3. The following 4 questions are related to the same economy.

In this economy, the consumption function is given by $C = 400 + 0.5(Y - T)$. Here Y is the GDP and T is Taxes. Investment is 100 units and government expenditure is 75 units.

- Prove that $MPC + MPS = 1$ for this economy.
- If government expenditure increases by 10 units and taxes increase by 8 units, to what proportion the GDP will change?
- If investment function changes to $I = 300 - 50r$. Where r is the rate of interest. Write down the IS curve equation and draw the IS curve for $r = 0, 1$, and 2 . (2 marks)
- Suppose the Money demand function is $(M/P)_d = Y - 60r$. The money supply is fixed at Rs 1000. Write down the LM curve equation and draw the LM curve for $r = 0, 1$, and 2 .